# Shivam Kumar Verma

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## Experience

**poiro.com** **Jul 2026 – Present**
*Fullstack Developer Intern* *Python, FastAPI, MongoDB, Redis, Celery, Next.js, TypeScript, Cloudflare R2*

- **AI Workflow Engine:** Built the node and group system behind the AI workflow builder: each workflow is a DAG of generation steps passing typed outputs to the next, and groups run nested steps as one sub-workflow.
- **Fail-Closed Provider Layer:** Integrated Pixabay, Unsplash and Pinterest behind one adapter contract (Pinterest vendors load-tested at **30 concurrent requests**) with org-scoped Redis caching, capping paid calls at **500 per org per day** and **5,000 globally**, counted only on cache misses.
- **Shared Asset Service:** Designed the org-scoped MongoDB data models and FastAPI backend for one asset service shared by **3 apps**, with AI provenance on every asset and pHash/dHash duplicate blocking at upload.

**klimb.io** **Nov 2025 – May 2026**
*Full Stack Developer Intern* *TypeScript, Express.js, PostgreSQL, Redis, AWS*

- **AI Usage Ledger:** Architected a PostgreSQL credit ledger enforced in middleware that records every use of **6+ AI features** per customer, replacing on/off feature flags and making usage-based billing possible.

## Projects

**DeepDoc | GitHub | *RAG, Next.js, TypeScript, Azure OpenAI, Pinecone, PostgreSQL***

- **RAG Pipeline:** Splits each PDF into section-aware chunks, searches only that document's vectors, rewrites follow-up questions from chat history, and packs deduplicated context into a token budget.
- **Evaluation Harness:** Created a **36-case** harness (multi-turn, Hinglish, prompt injection) that separates indexing, retrieval and generation failures; it traced empty answers to a field ingestion never wrote, after prompt changes had failed.

**Hireflow | GitHub | *Node.js, TypeScript, BullMQ, Redis, PostgreSQL, Playwright, Gemini***

- **Grounded Extraction:** Built an agentic pipeline that ingests listings from **4 sources**, extracts stack and seniority with Gemini, rewrites resume bullets with GPT-4o-mini, and runs a validator that rejects any skill not in the real profile, so the model cannot invent one.
- **Decoupled Stages:** Split ingestion, AI processing and Playwright automation onto separate BullMQ queues so a bot-detection block in one stage never stalls the others; SHA-256 fingerprints dedup listings across sources.

**Storix | GitHub | Live | *FastAPI, Celery, Redis, PostgreSQL, AWS S3***

- **Async Ingestion Pipeline:** Developed a PDF ingestion pipeline where FastAPI accepts the upload, **Celery** workers extract off the request path, and Redis pub/sub streams each stage's status to the client.

**CodeSM | GitHub | Live | *Node.js, BullMQ, Redis, PostgreSQL, Docker***

- **Sandboxed Execution:** Implemented isolated grading for **6 languages** on BullMQ workers, with Docker limits of **256 MB**, **32 PIDs** and **0.5 CPU** per run and no network access.

## Skills

**AI Engineering:** document extraction, RAG, embeddings and vector search, LLM evaluation, agent workflows, Azure OpenAI, Gemini
**Languages:** Python, TypeScript, SQL, C++
**Backend & Frontend:** FastAPI, Node.js, Express.js, REST APIs, WebSockets, Celery, BullMQ, React, Next.js
**Data & Infrastructure:** PostgreSQL, MongoDB, Redis, Pinecone, Docker, Kubernetes, AWS (EC2, S3)

## Education

**Indian Institute of Information Technology Nagpur** **Nov 2022 – May 2026**
*B.Tech in Computer Science and Engineering* *Nagpur, Maharashtra*

## Achievements

- **Competitive Programming: Top 5% globally** on LeetCode (**1800+ rating**, 800+ problems solved) · **3-star** on CodeChef (1700+) · Pupil on Codeforces.
- **Hackathon: 3rd place** at Tantrafiesta 2024, IIIT Nagpur.